The citemsp Package

Per-Key Citation Locators for Biblatex

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Abstract

The `citemsp` package extends `biblatex` with a single user-facing command, `\citemsp`, that attaches section (§), paragraph (¶), and prefix-based locators directly to numeric citation labels. An extensible prefix registry provides six built-in types — equation (eq.), definition (≜), figure (fig.), footnote (†), page (pp.), and table (⊞) — and users can register custom prefixes with `\citemspprefix`. Standard `biblatex` provides locator fields through its postnote mechanism, but these cannot display stacked locator pairs per key in a compact inline form. The `citemsp` package solves this by parsing a slash-delimited syntax, rendering each locator as a superscript/subscript pair on the corresponding citation number, producing output such as [**wald1984**, **mtw1973**, **hawking1974**]. The package is compatible with all `biblatex` numeric styles and requires only `biblatex`, `graphicx`, and `amssymb`.

Contents

1 Motivation

The `biblatex` package offers a locator interface through its postnote field. A user can write `\autocite[§3.2]{wald1984}` to append a free-form locator after the citation number. For multi-key citations, `\autocites` accepts per-key pre- and post-notes:

```
\autocites[§3.2]{wald1984}[¶2]{hawking1974}
```

This approach has two limitations when writing a scientific paper:

1. **Verbose syntax.** Citing three sources with locators requires nested optional arguments, which is cumbersome in source files.
2. **No typed locators.** There is no compact notation for referencing specific sections, paragraphs equations, figures etc.

The `citemsp` package renders locators as a superscript/subscript pair attached to the citation number, supports section and paragraph locators by default plus seven additional types via an extensible prefix registry, and accepts all keys with their locators in a single comma-separated argument. It also allows for the user to define their own personal locators.

2 Installation

2.1 Per-project

Copy `citemsp.sty` into the same directory as your main `.tex` file.

2.2 System-wide (local texmf)

```
mkdir -p ~/texmf/tex/latex/citemsp
cp citemsp.sty ~/texmf/tex/latex/citemsp/
texhash ~/texmf
```

3 Usage

3.1 Loading

The package must be loaded *after* `biblatex`:

```
\usepackage[style=numeric, backend=biber]{biblatex}
\usepackage{citemsp}
```

Loading `citemsp` before `biblatex` will produce a package error.

3.2 Syntax

The `\citemsp` command accepts a comma-separated list of entries. Each entry has the form `key`, `key/loc1`, or `key/loc1/loc2`. By default, the first slot renders as § (section) and the second as ¶ (paragraph). A single-letter prefix changes the locator type:

`\citemsp{key}`	$\longrightarrow$	[**wald1984**]
`\citemsp{key/section}`	$\longrightarrow$	[**wald1984**]
`\citemsp{key/section/paragraph}`	$\longrightarrow$	[**wald1984**]
`\citemsp{key/section/e4.3}`	$\longrightarrow$	[**wald1984**]

The built-in prefixes are: `d` for definition (≜), `e` for equation (eq.), `f` for figure (fig.), `n` for footnote (†), `p` for page (pp.), and `t` for table (⊞). No prefix defaults to § or ¶ depending on slot position. When both locators are present, the first appears as a superscript and the second as a subscript. All are found in the following table:

Prefix	Locator	Render	Output
`d`	Definition	≜	[**wald1984**]
`e`	Equation	eq.	[**wald1984**]
`f`	Figure	fig.	[**wald1984**]
`n`	Footnote	†	[**hawking1974**]
`p`	Page	pp.	[**wald1984**]
`t`	Table	⊞	[**mtw1973**]

Table 1: Registry: each single-letter prefix maps to a typographical symbol rendered before the locator number.

Users can register custom prefixes with `\citemspprefix`:

```
\citemspprefix{w}{w.}   % register "w" prefix rendering as "w."
```

3.3 Configuration

The locator glyph size is controlled by `\citemspscale` (default 0.35). Redefine it after loading:

```
\renewcommand{\citemspscale}{0.4}
```

4 Comparison with Standard Biblatex

5 Examples

5.1 Single source

```
The Schwarzschild metric~\citemsp{wald1984/6.1/2} describes
spherically symmetric vacuum solutions.
```

Output: The Schwarzschild metric [**wald1984**] describes spherically symmetric vacuum solutions.

5.2 Multiple sources with mixed locators

```
General covariance~\citemsp{einstein1915, wald1984/4,
  hawking1974/1/2} is the foundation of general relativity.
```

Output: General covariance [**einstein1915**, **wald1984**, **hawking1974**] is the foundation of general relativity.

5.3 Italic context

The package detects italic font shape and adjusts kerning automatically:

```
\textit{Hawking radiation~\citemsp{hawking1974/1/2} implies
black holes are not truly black.}
```

Output: *Hawking radiation [***hawking1974***] implies black holes are not truly black.*

5.4 Equation locator

Use the `e` prefix to reference a specific equation:

```
The field equations~\citemsp{wald1984/3.2/e4.3} can be derived
from a variational principle.
```

Output: The field equations [**wald1984**] can be derived from a variational principle.

5.5 Mixed locator types

All locator types can be freely mixed in a single call:

```
See~\citemsp{wald1984/6.1/2, mtw1973/p520, hawking1974/1/e4.3}
for further details.
```

Output: See [**wald1984**, **mtw1973**, **hawking1974**] for further details.

5.6 Additional locator types

The prefix registry provides six built-in types beyond the default §/¶:

```
\citemsp{wald1984/3.2/d5}          % definition
\citemsp{wald1984/6.1/f3}          % figure
\citemsp{hawking1974/1/n7}         % footnote
\citemsp{wald1984/p42}             % page
\citemsp{mtw1973/8.4/t2}          % table
```

Output: [**wald1984**], [**wald1984**], [**hawking1974**], [**wald1984**], [**mtw1973**].

6 Implementation

1. **Locator scaling.** `\citemspscale` (default 0.35) and `\citesize` control the size of locator glyphs via `\scalebox` from `graphicx`.

2. **Citation rendering.** A custom biblatex cite command, `\citelinksp`, declared via `\DeclareCiteCommand`, prints the hyperlinked citation number and conditionally attaches superscript/subscript locators through the dispatcher. When `hyperref` draws bordered link boxes (`colorlinks=false`), a horizontal kern prevents overlap.

3. **Entry parsing.** The user-facing `\citemsp` uses `\forcsvlist` (from `etoolbox`, loaded by `biblatex`) to iterate over comma-separated entries. Each entry is split at forward slashes using a delimited `\def` pattern, extracting the key and up to two locator slots before calling the renderer.

7 Conventions

- **Section numbering:** use the source's native format (e.g., 3.2.1, IV, A.1).
- **Paragraph counting:** count paragraphs from the start of the referenced section.
- **Prefix locators:** ideally use a single-letter prefix for typed locators: `d` (definition), `e` (equation), `f` (figure), `n` (footnote), `p` (page), `t` (table). E.g., `key/3.2/e4.3` for an equation, `key/p42` for a page.
- **Subsection references:** use the full path (e.g., `key/3.2.1`), not a relative number.
- **Plain citations:** `\citemsp{key}` with no slashes produces a standard numeric citation, so the command can replace `\cite` if desired.

8 Requirements

- LaTeX2e (2020/10/01 or later)
- `biblatex` (any numeric style) with Biber backend
- `graphicx`
- `amssymb` (for $\square$, $\triangleq$, $\boxplus$, † locator glyphs)

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